Lakshya JEE 2.0 (2024)

Limit, Continuity and Differentiability

DPP-05

1. For what value of k is the function $f(x)=\begin{cases}\dfrac{x^2-9}{x-3}, & \text{when } x\neq 3\\ k, & \text{when } x=3\end{cases}$ is continuous at $x = 3$?

2. Find the value of k for which the function $f(x)=\begin{cases}\dfrac{k\cos x}{\pi-2x}, & \text{if } x\neq\dfrac{\pi}{2}\\ 3, & \text{if } x=\dfrac{\pi}{2}\end{cases}$ is continuous at $x=\dfrac{\pi}{2}$.

3. Consider the function $f(x)$ = $\begin{cases}2x^2+3, & if\ x\leq 3\\ 3x+B, & if\ x>3\end{cases}$ find a value of B such that $f(x)$ is continuous at x = 3

(1) 6 (2) 9

(3) 12 (4) 15

4. Find all values of 'a' such that the function $f(x)$ = $\begin{cases}x^2+2x, & if\ x<a\\ -1, & if\ x\geq a\end{cases}$ is continuous everywhere

(1) a = –1 only (2) a = –2 only

(3) a = –1 and a = 1 (4) a = –2 and a = 2

5. Suppose $f(t)$ = $\begin{cases}Bt & if\ t\leq 3\\ 5 & if\ t>3\end{cases}$ find a value of 'B' such that the function $f(t)$ is continuous for all 't'

(1) 3/5 (2) 4/5

(3) 5/3 (4) 5/4

6. Let $f(x)$ = . $\dfrac{1-\tan x}{4x-\pi}, x\neq\dfrac{\pi}{4}, x\in\left[0,\dfrac{\pi}{2}\right]$.. If $f(x)$ is continuous in $\left[0,\dfrac{\pi}{2}\right]$ then $f\left(\dfrac{\pi}{4}\right)$ is

(1) –1 (2) $\dfrac{1}{2}$

(3) $-\dfrac{1}{2}$ (4) 1

7. The value of p and q for which the function

$$f(x) = \begin{cases}\dfrac{\sin(p+1)x+\sin x}{x}, & x<0\\ q, & x=0\\ \dfrac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}}, & x>0\end{cases}$$

Continuous for all x in R are

(1) $p=\dfrac{5}{2}$, $q=\dfrac{1}{2}$ (2) $p=\dfrac{-3}{2}$, $q=\dfrac{1}{2}$

(3) $p=\dfrac{1}{2}$, $q=\dfrac{3}{2}$ (4) $p=\dfrac{1}{2}$, $q=\dfrac{-3}{2}$

8. Find the value of λ for which $f(x)=\begin{cases}\dfrac{x^2-2x-3}{x+1}, & when\ x\neq -1\\ \lambda, & when\ x=-1\end{cases}$ is continuous at x = –1

9. If $f(x)=\begin{cases}\dfrac{2^{x+2}-16}{4^x-16}, & x\neq 2\\ k, & x=2\end{cases}$ is continuous at x = 2, then find k.

10. Find the values of a for which the function f, defined as $f(x)=\begin{cases}a\sin\dfrac{\pi}{2}(x+1), & x\leq 0\\ \dfrac{\tan x-\sin x}{x^3}, & x>0\end{cases}$ is continuous at x=0.

11. Which of the following is true about

$$f(x)=\begin{cases}\dfrac{(x-2)}{|x-2|}\left(\dfrac{x^2-1}{x^2+1}\right), & x\neq 2\\[2mm] \dfrac{3}{5}, & x=2\end{cases}$$

(1) $f(x)$ is continuous at $x = 2$

(2) $f(x)$ has removable discontinuity at $x = 2$.

(3) $f(x)$ has non-removable discontinuity at $x = 2$

(4) Discontinuity at $x = 2$ can be removed by redefining the function at $x = 2$.

12. If the function

$$f(x)=\begin{cases}\dfrac{1}{x}\log_e\left(\dfrac{1+\frac{x}{a}}{1-\frac{x}{b}}\right), & x<0\\[2mm] \dfrac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1}, & x>0\\[2mm] K, & x=0\end{cases}$$

Continuous at $x = 0$, then $\frac{1}{a}+\frac{1}{b}+\frac{4}{K}$ is equal to:

(1) –5 (2) 5

(3) –4 (4) 4

13. Let a, b $\in$ R, b $\in$ 0, Define a function

$$f(x)=\begin{cases}a\sin\dfrac{\pi}{2}(x-1), & for\ x\le 0\\[2mm] \dfrac{\tan 2x-\sin 2x}{bx^3}, & for\ x>0.\end{cases}$$. If ‘f’ is

continuous at $x = 0$, then ab is equal to:

(1) 4 (2) –4

(3) 14 (4) –14

14. Let $f: R\to R$ be defined as

$$f(x)=\begin{cases}\dfrac{\sin(a+1)x+\sin 2x}{2x}, & if\ \ x<0\\[2mm] \mathbf{b}, & if\ \ x=0\\[2mm] \dfrac{\sqrt{x+bx^3}-\sqrt{x}}{bx^{5/2}}, & if\ \ x>0\end{cases}$$

If f is continuous at $x = 0$, then the value of $a + b$ is

(1) –3

(2) – 2

(3) $\dfrac{-5}{2}$

(4) $\dfrac{-3}{2}$

15. Let $f:\left(-\frac{\pi}{4},\frac{\pi}{4}\right)\to R$ be defined as

$$f(x)=\begin{cases}(1+|\sin x|)^{\frac{3a}{|\sin x|}}, & \dfrac{-\pi}{4}<x<0\\[2mm] \mathbf{b}, & x=0\\[2mm] e^{\frac{\cot 4x}{\cot 2x}}, & 0<x<\dfrac{\pi}{4}\end{cases}$$

If f is continuous at $x = 0$, then the value of $6a+b^2$ is

(1) 1 – e (2) e – 1

(3) 1 + e (4) e

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. **k = 6**
2. **k = 6**
3. **(3)**
4. **(1)**
5. **(3)**
6. **(3)**
7. **(2)**
8. **$\lambda = -4$**
9. $k = \frac{1}{2}$
10. $a = \frac{1}{2}$
11. **(3)**
12. **(1)**
13. **(2)**
14. **(4)**
15. **(3)**

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